Leonardo Di Tommaso

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Research focus: satellite remote sensing, radiative transfer, and ML-based bias correction for greenhouse gas retrievals

Education

Harvard University Cambridge, MA
AB in Physics, GPA 3.6 *Expected May 2028*

Relevant Coursework: Graduate Atmospheric Chemistry and Physics; Mathematical Methods for Physics; Wave Phenomena; Quantum Mechanics 1; Computational Problem Solving with Python; Linear Algebra and Differential Equations; Data Analysis and Statistical Inference in the Earth and Environmental Sciences

Liceo Scientifico E. Bérard Aosta, Italy
GPA: 8.92 Italian school system, top 5% of class, SAT: 1510, TOEFL: 104/120 *July 2024*

Research Experience

California Institute of Technology Pasadena, CA, USA
Undergraduate Researcher (Advisors: Prof. C. Frankenberg, Prof. S. Wofsy) *June 2026 - Present*
Accurate Greenhouse Gas Retrievals: Leveraging Surface Heterogeneity for Correcting Light-Path Variations

- Built a three-stage ML and physical-retrieval pipeline (regression and neural networks in Julia/Flux.jl) on vSmartMOM radiative transfer simulations to correct aerosol-scattering bias in MethaneSAT CO_2 retrievals.
- Generated 2,000 synthetic satellite scenes across 5 aerosol types and geometries, reducing the median aerosol-induced CO_2 bias from 9.89 ppm to 0.34 ppm (0.079%) on 88 held-out test scenes, meeting the 0.25% accuracy threshold for regional CO_2 flux quantification.
- Presented results in a presentation; currently extending the method to real MethaneSAT observations toward an operational bias-correction algorithm for high-resolution regional CO_2 flux retrievals.

MethaneSAT XCO_2 Correction Development, Harvard University Cambridge, MA, USA
Undergraduate Researcher (Advisor: Prof. S. Wofsy) *September 2025 - May 2026*

- Built an automated Python/R data pipeline correcting MethaneSAT XCO_2 retrievals for scattering and orography biases, integrating MethaneSAT and VIIRS observations with 6km spatial alignment and $\pm$90-minute temporal matching across 100+ scenes.
- Implemented a two-stream Eddington radiative transfer model and trained ML emulators (random forest, MLP; $R^2 > 0.99$) to correct aerosol-induced retrieval bias at pixel scale.
- Developed a statistical correction framework using aerosol optical properties and viewing geometry, applying regression with multicollinearity diagnostics (VIF) and AICc model selection to remove systematic biases from Earth observation data.

Harvard Summer Research Village (PRISE), Harvard University Cambridge, MA, USA
Undergraduate Researcher (Advisor: Prof. S. Wofsy) *June 2025 - August 2025*

- Built automated R pipelines for NetCDF data extraction, geospatial alignment, and raster-based analysis of satellite atmospheric data.
- Applied robust linear models to quantify systematic biases in ΔPressure and XCO_2 caused by atmospheric scattering; presented results in a research poster.

Italian EPA (ARPA) Aosta, Italy
Intern/Researcher *July 2023 - Nov. 2023*

- Researched UV dosimeter accuracy and recalibrated instrumentation via multi-parameter analysis in MapleSoft; published findings (DOI: 10.5281/zenodo.10071501).

Leadership & Activities

Opson Society, Harvard University Cambridge, MA, USA
Founder & President *Present*

- Founded and lead a student organization of 300+ members dedicated to cuisine, organizing intercultural cooking workshops and dinner receptions.

Italian Regional Board for Student Representation (Consulta Studentesca) Aosta, Italy
Regional President of Student Representatives *Oct 2023 - Oct 2024*

- Advised the Italian Ministry of Education on national policy reforms and, as a G7 Young for Education delegate, co-authored mental-health and educational-inclusion proposals adopted at the Tivoli 2024 summit.
- Organized career orientation events and wellness campaigns (substance abuse awareness, financial literacy) across regional schools.

Skills & Interests

Programming & Software: Python, Julia, R (terra, raster, ncdf4, tidyverse, ggplot2), Git, LaTeX, Maple

Machine Learning & Data Science: Supervised ML (random forest, MLP), regression, spatial analysis, AICc/VIF model selection

Domain: Satellite remote sensing, radiative transfer | **Languages:** English (fluent), Italian (native), French